Adrian Lopez Lopez – Mechanical Engineering Portfolio

Engineering Projects, Certifications, and Technical Experience

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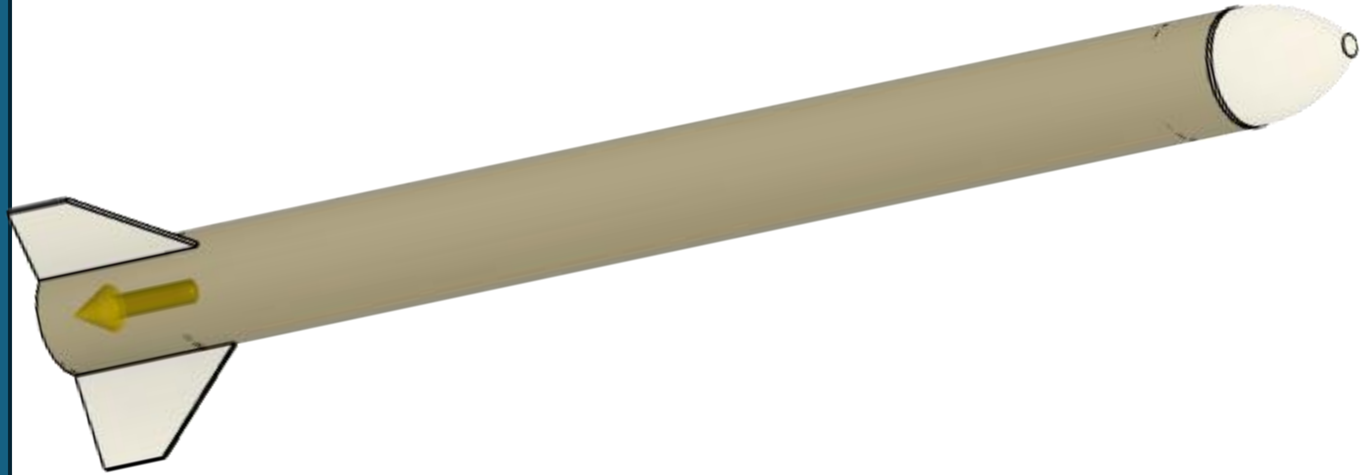
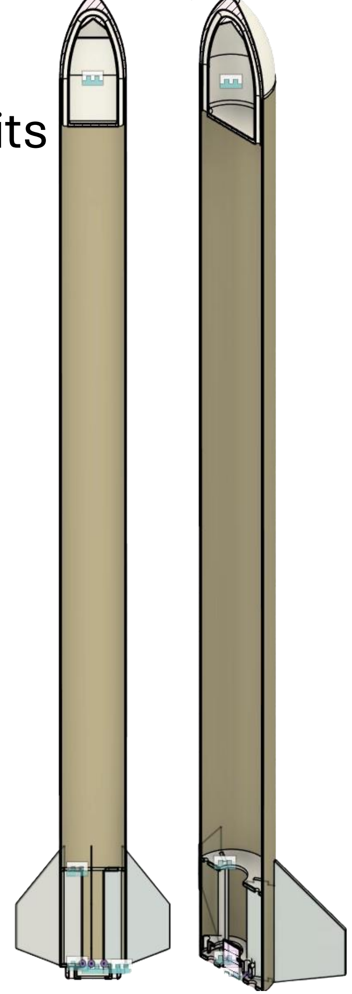
Prepared for Recruiters & Hiring Managers

Highlander Space Program (L1 Certification) — Fusion 360 CAD, Simulation & Build

Completed: August, 2026

- I **modeled** and **built** a complete rocket in **Fusion 360**, **designing** an **ogive nose cone, airframe, centering rings, fins, and rail guides**. I **simulated** its performance in **OpenRocket**, achieving a **2,900 ft apogee**, **660 ft/s max velocity**, and **stable flight**. I **manufactured** the rocket using epoxy and **applied** cellulose fiber insulation ("dog barf") for **thermal protection**, shielding the parachute from hot ejection gases using heat-transfer principles. The total build took roughly 4 hours plus curing time.



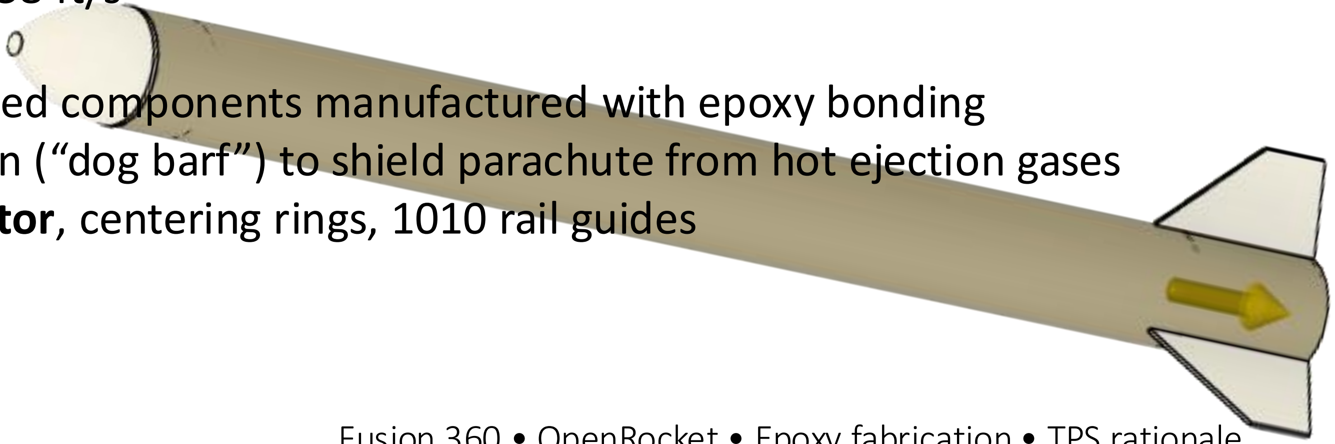


Fusion 360 • OpenRocket • Epoxy fabrication • TPS rationale

L1 Certification- Overview, Specifications & Materials

Completed: August, 2026

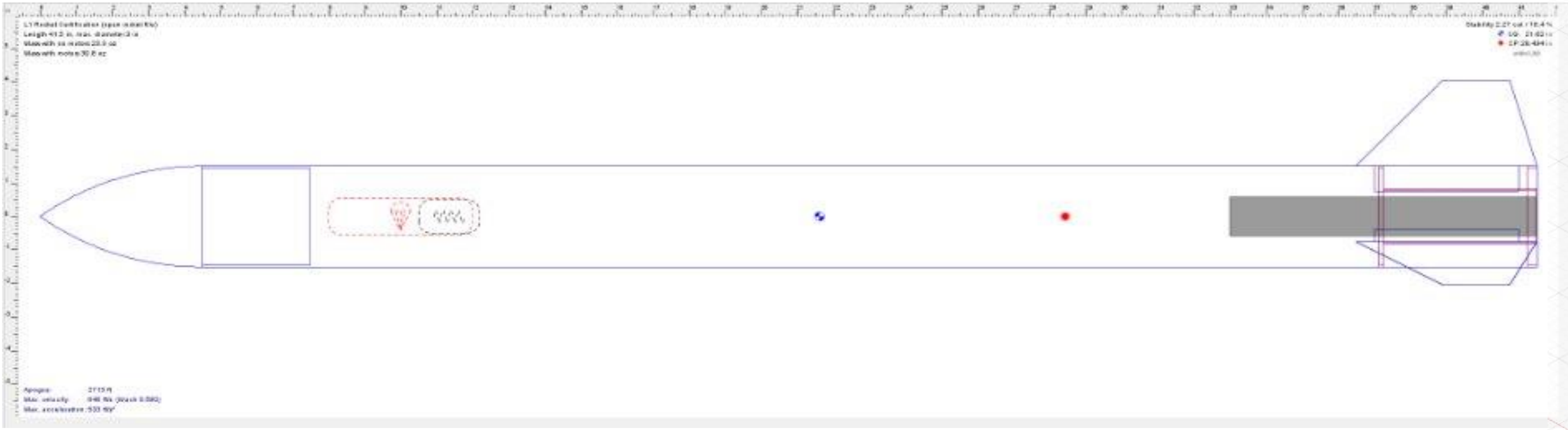
- **Overview:**
  - **Designed, simulated, and built** a full-scale model rocket to dem**onstrate proficiency in CAD**, **structural analysis**, and **manufacturing**. Focused on **aerodynamic stability, thermal protection, and precision assembly** to achieve a **safe** and controlled flight.
- **Specifications:**
  - Length: 41.5 in | Diameter: 3 in
  - Weight: ~2 lb | Max velocity: 660 ft/s | Apogee: 2,900 ft
  - Acceleration: 552 ft/s^2 | Flight time: 123 s | Mach 0.580
  - Stability: 2.37 cal (CG 21.62 in, CP 28.4 in)
  - Ejection delay: 9 s | Deployment velocity: 1.58 ft/s
- **Materials:**
  - Airframe, nose cone, fins: Fusion 360-modeled components manufactured with epoxy bonding
  - **Thermal protection:** cellulose fiber insulation ("dog barf") to shield parachute from hot ejection gases
  - **Structural hardware: Aerotech H135-14 motor**, centering rings, 1010 rail guides

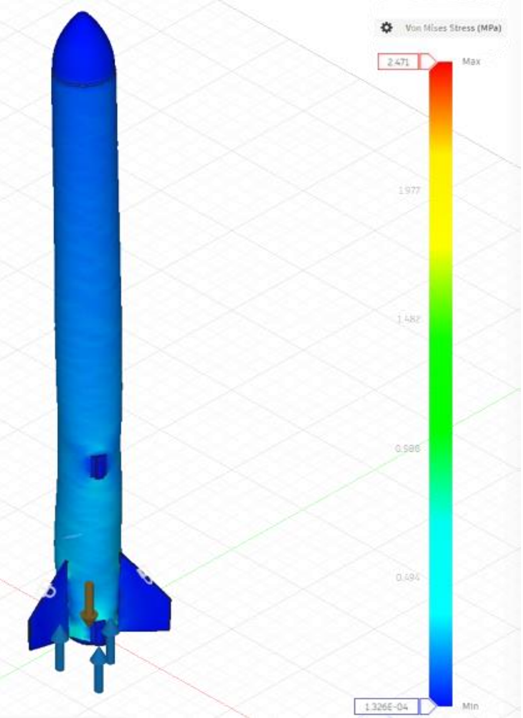


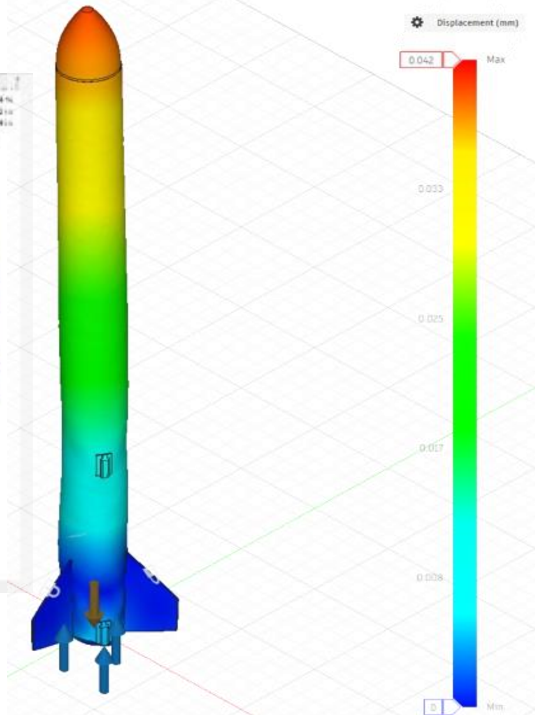
Fusion 360 • OpenRocket • Epoxy fabrication • TPS rationale

Open rocket, CAD & FEA Analysis

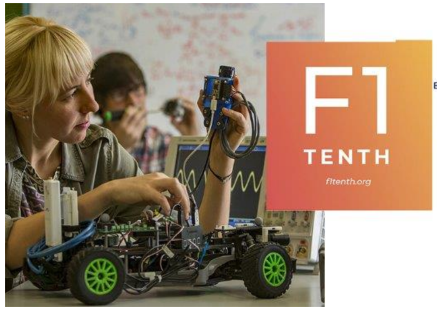
Name	Configuration	Velocity off rod	Apogee	Velocity at deploy...	Optimum delay	Max. velocity	Max. acceleration	Time to apogee	Flight time	Ground hit velocity
Simulation 1	[HP-H135W-9.6]	56.1 ft/s	2713 ft	14.2 ft/s	9.2 s	646 ft/s	533 ft/s²	11.2 s	154 s	18.7 ft/s
Simulation 2	[HP-H135W-9.6]	145 ft/s	2857 ft	3.76 ft/s	9.55 s	657 ft/s	545 ft/s²	11.6 s	155 s	19.5 ft/s
Simulation 3	[HP-H135W-9.6]	145 ft/s	2857 ft	2.88 ft/s	9.51 s	656 ft/s	545 ft/s²	11.6 s	155 s	19.5 ft/s
Simulation 4	[HP-H135W-9.6]	146 ft/s	2828 ft	1.58 ft/s	9.41 s	660 ft/s	552 ft/s²	11.5 s	123 s	25 ft/s
Simulation 5	[HP-H135W-9.6]									
Simulation 6	[HP-H135W-9.6]	147 ft/s	2861 ft	3.26 ft/s	9.48 s	663 ft/s	554 ft/s²	11.5 s	124 s	25 ft/s
Simulation 7	[HP-H135W-9.6]	147 ft/s	2863 ft	3.79 ft/s	9.48 s	663 ft/s	554 ft/s²	11.5 s	124 s	25 ft/s
Simulation 8	[HP-H135W-9.6]	145 ft/s	2856 ft	4.34 ft/s	9.52 s	657 ft/s	545 ft/s²	11.6 s	155 s	19.5 ft/s
Simulation 9	[No motors]									
Simulation 10	[No motors]									
Simulation 11	[HP-H135W-9.6]	145 ft/s	2856 ft	3.28 ft/s	9.55 s	657 ft/s	545 ft/s²	11.6 s	155 s	19.5 ft/s

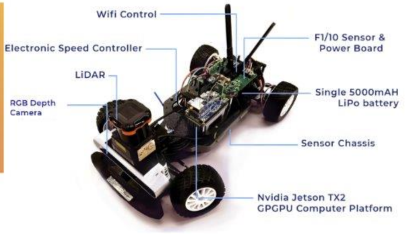


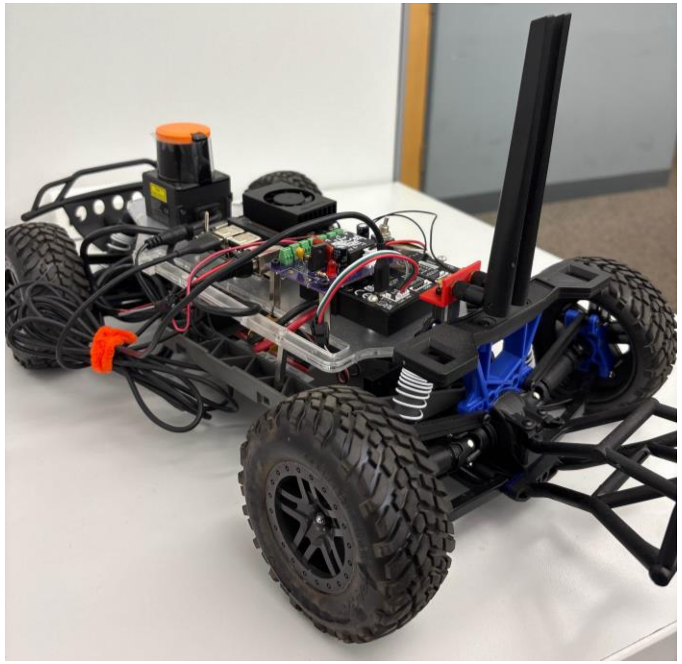




F1Tenth — 2D LiDAR Autonomy on Jetson Nano with VESC Mk VI





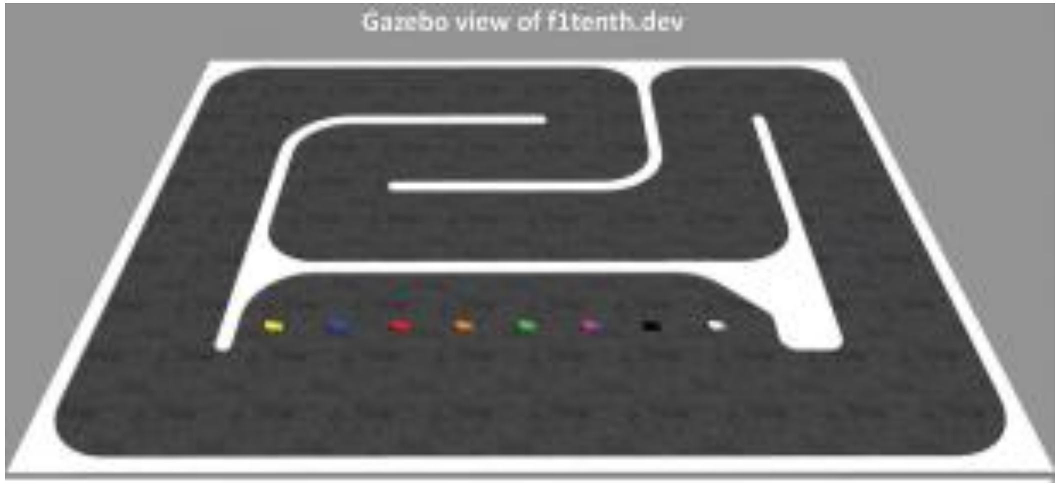


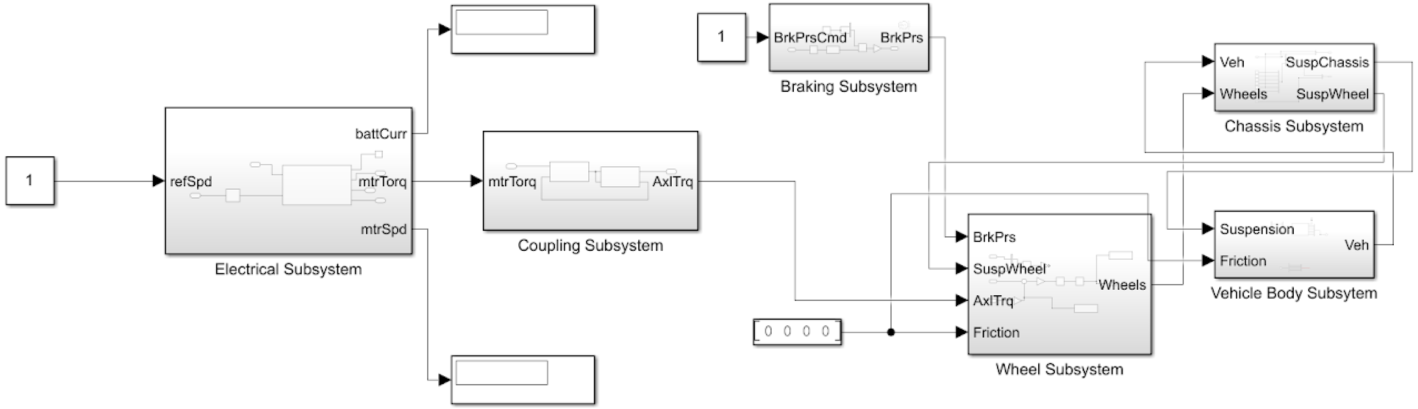
- **Overview:**
  - **Developed a 4-motor autonomous vehicle platform integrating Arduino and Jetson Nano for 2D LiDAR-based navigation. Implemented real-time sensor fusion and PID motor control for smooth, responsive driving, preparing the system for the May F1Tenth competition. Designed chassis, mounts, and brackets in Siemens NX for optimal robustness and manufacturability; simulated vehicle dynamics in MATLAB/Simulink to test ABS, throttle, and regenerative braking virtually.**
- **Specifications:**
  - **Sensors**: MPU6050 IMU, rotary encoders, INA219 current sensing
  - **Control**: PID for throttle and steering, real-time serial communication between MCU and Jetson Nano
  - **Software**: RROS2 for sensor fusion and autonomy pipeline; **MATLAB/Simulink for HIL simulations and virtual testing**
  - **Competition car setup:** VESC Mk VI configured for regenerative braking, smooth deceleration without skidding, motor/servo calibration, and PID tuning
- **Materials & Build:**
  - Base platform designed to mount LiDAR, Jetson Nano, VESC Mk VI, 3-cell 6000 mAh battery, power board, and Wi-Fi antennas
  - Drivetrain isolated from electronics for robustness
  - Physical integration and calibration of all mechanical and electrical components

ROS2 • PID • VESC • Mech-E integration

F1Tenth — 2D LiDAR Autonomy on Jetson Nano with VESC Mk VI

- I **developed** a 4-motor autonomous vehicle platform for the **F1Tenth competition**. I **modeled** and **optimized** the chassis, mounts, and components in **Siemens NX** for **robustness** and **manufacturability**. I **implemented** real-time sensor fusion and PID motor control using **ROS2**, and **simulated** braking, throttle, and **regenerative** systems in **MATLAB/Simulink**, **reducing physical testing time** and component wear. I **delivered** a fully calibrated, high-performance prototype ready for competition.

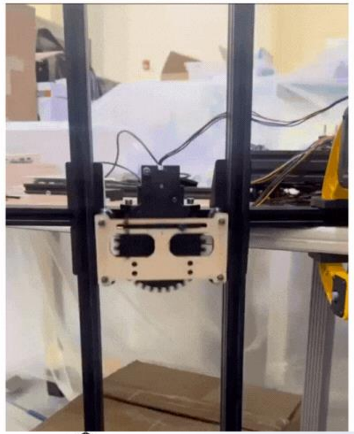


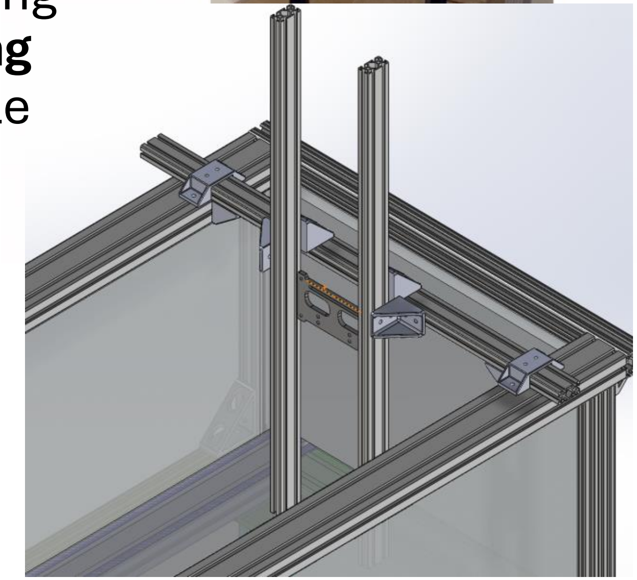


ROS2 • PID • VESC • Mech-E integration

Lab-Scale Wave Generator — Scotch Yoke Drive, Sizing & Validation

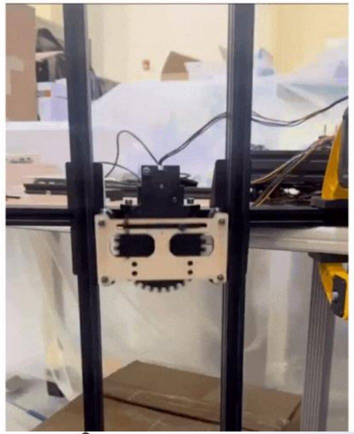
- I **designed** and built a lab-scale wave generator using a **Scotch yoke mechanism** to produce ±3 in sinusoidal waves at 1 Hz. I powered the system with a **NEMA 17 stepper motor** and 3:1 gear reduction, ensuring torque exceeded **hydrodynamic requirements**. I sized the waves using linear theory ($\lambda \approx 1.21$ m, $S \approx 0.30$) and **prototyped** the mechanism with **laser-cut components**, adding a variable-radius crank for amplitude tuning and **structural reinforcements** for **reliability**. I conducted **dry-run testing** to confirm **smooth**, **repeatable** motion and **stability**, supporting lab-scale vortex dynamics and **Strouhal number experiments**.

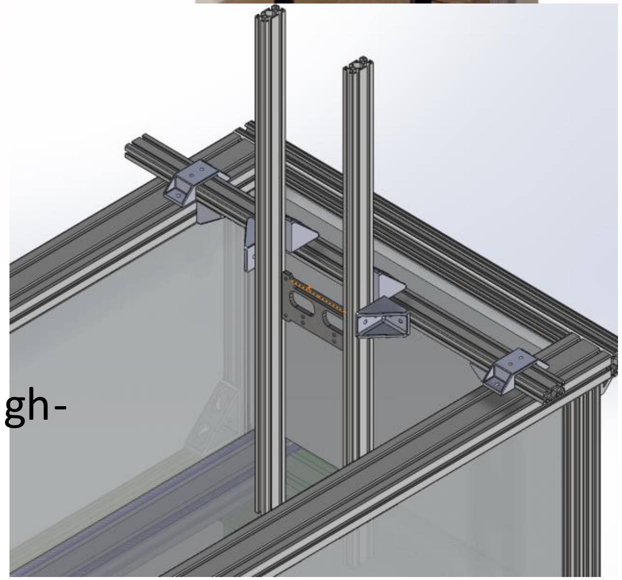




Mechanism design • Fluids • Sizing calcs • Rapid prototyping

Lab-Scale Wave Generator — Scotch Yoke Drive, Sizing & Validation

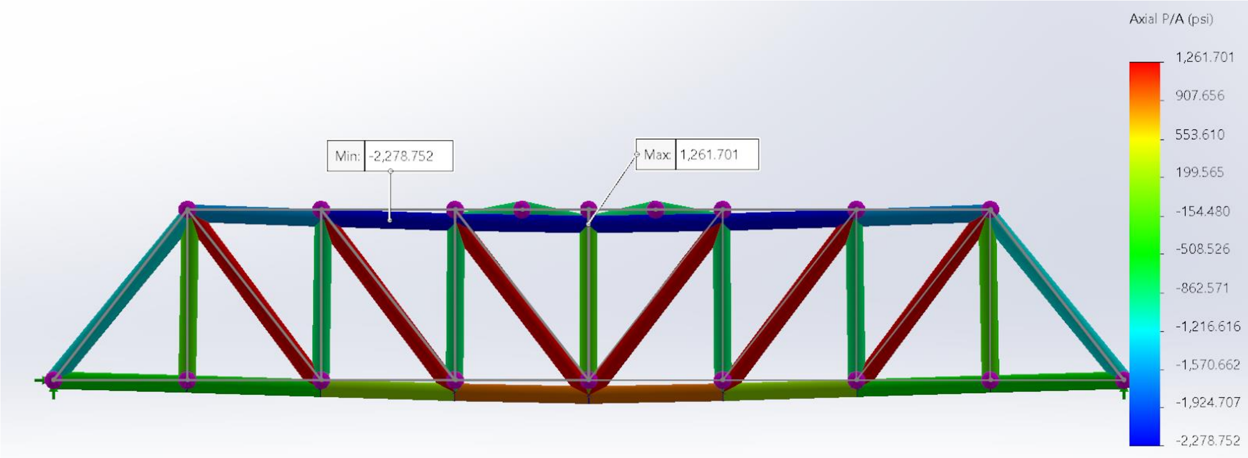


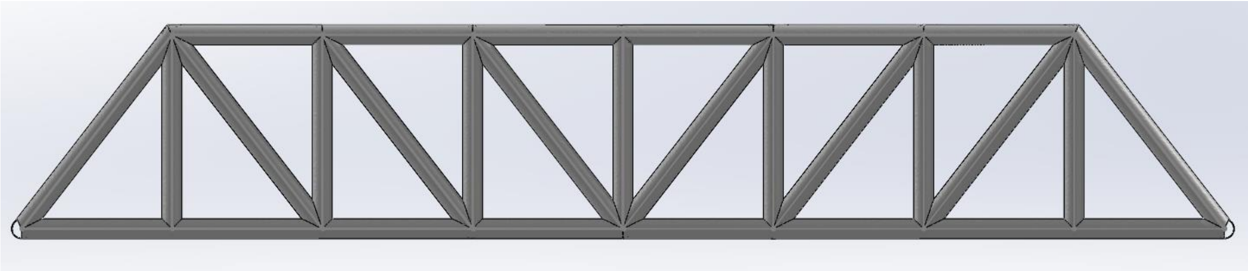


- **Overview:**
  - **Designed and validated a lab-scale wave generator capable of producing continuous, unidirectional sinusoidal waves for bio-inspired propulsion and renewable energy research. Transitioned from a rack-and-pinion to a Scotch yoke mechanism for smooth, low-reflection motion.**
- **Specifications:**
  - Stroke: ±3 in at 1 Hz (60 RPM)
  - Torque: ~135 N·cm available (required ~37.5 N·cm)
  - Wave parameters: $\lambda \approx 1.21$ m, amplitude-to-wavelength ratio $S \approx 0.30$
  - Drive: NEMA 17 stepper (1.5 A/phase, 45 N·cm) with 3:1 gear reduction
- **Materials & Build:**
  - Laser-cut wood prototypes for rapid iteration, later to transition to aluminum or high-impact polymers
  - Variable-radius crank arms for amplitude tuning
  - Precision slot–pin alignment and structural reinforcements
  - Minimalist Arduino-based driver for reliable motion, with future GUI integration planned

Mechanism design • Fluids • Sizing calcs • Rapid prototyping

FEA-Driven Optimization — 25-Straw Pratt Truss Bridge





- **Led** a **5-person** team; role assignment (materials tracking, cutting, assembly, data/presentation).
- **Process control**: designed **3D-printed PLA molds (5 mm)** using **SolidWorks Cavity** to ensure consistent geometry & reduce human error.
- **FEA insight**: mapped **principal compressive stress**; straws are **weakest in compression** → predicted failure site. *(Euler buckling form used conceptually:*

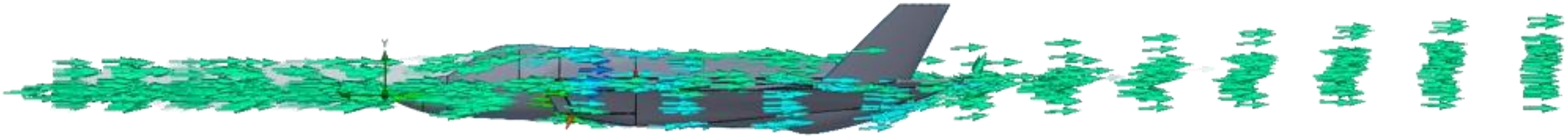
$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

  Iteration (18 bridges): added **top diagonals** between side trusses; **removed bottom horizontals** (unnecessary).
- **Prediction vs test**: **6,432 g** predicted; **19 g** bridge → **efficiency 336**.
- **Result**: **6,500 g** peak load on professor's fish-scale rig (personal bucket-and-water tests matched failure mode).

FEA • SolidWorks • Rapid iteration • Test correlation

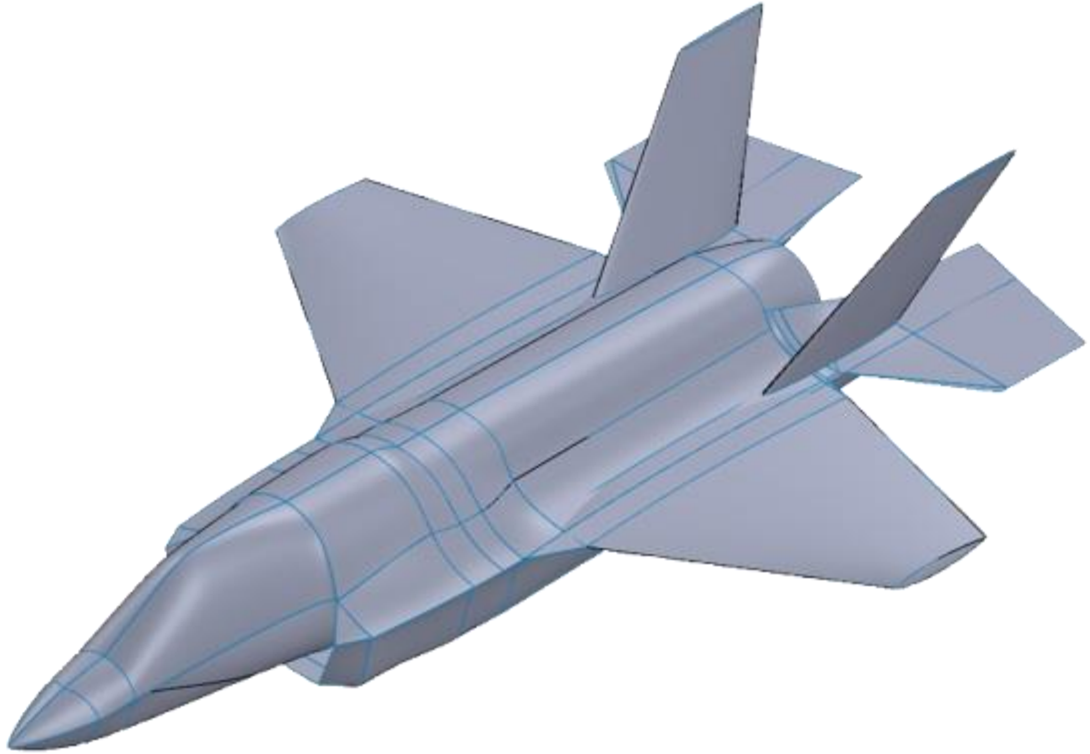
F-35B CAD Replication — 3D Sketching, Splines & Lofts

- I **replicated** an F-35B in **SolidWorks**, carefully correcting reference images and using **3D sketches, splines, and lofts** to model **complex fuselage**, **wing**, and **engine geometries**. I iteratively refined the design over ~1 month to **maximize accuracy** and **mirrored** the body to create a f**ull symmetric model.** External 3D renders guided detailing of challenging areas, and I plan to extend the model with **swivel/VTOL actuation** for a potential RC flight-capable variant.



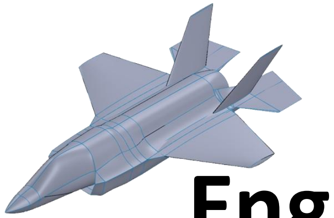
Solidworks /Surfacing • Reverse engineering • Parametric CAD

F-35B CAD Replication — 3D Sketching, Splines & Lofts

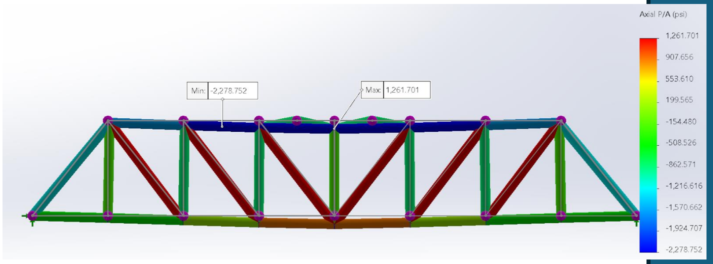


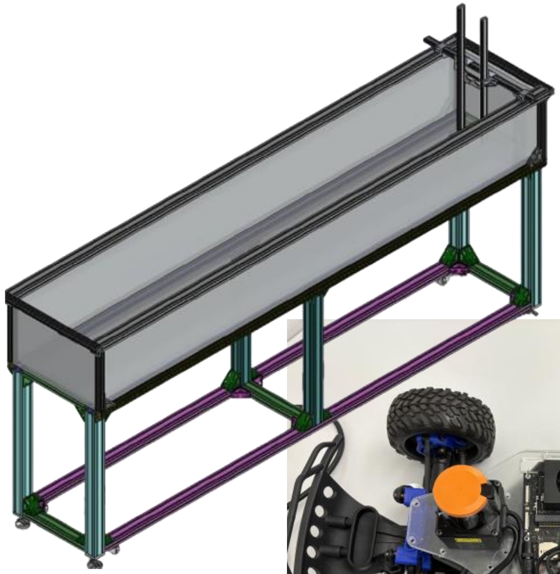
- **Overview:**
  - **Replicated** a full-scale F-35B jet in **SolidWorks**, **emphasizing** high-fidelity modeling for **complex geometries** and **accurate dimensional** representation.
- **Specifications:**
  - **Modeled fuselage**, **wings**, and **engine** regions using **3D sketches, splines, and lofts**; **mirrored** half the body to achieve symmetry. Iterative troubleshooting over ~1 month improved dimensional accuracy.
- **Materials/Tools:**
  - **SolidWorks CAD** software, **reference** photos corrected in **Photoshop**, external 3D renders for complex components.

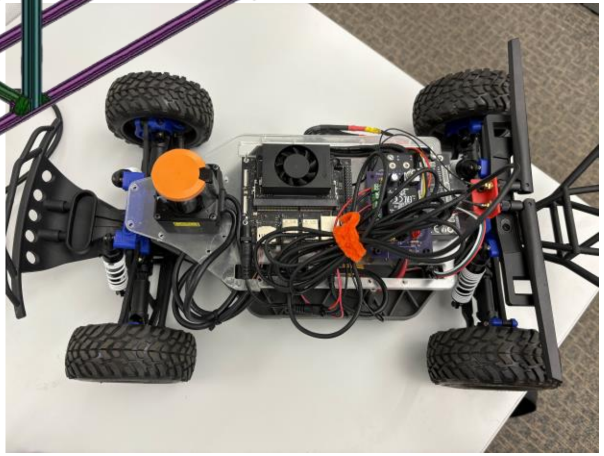
Solidworks /Surfacing • Reverse engineering • Parametric CAD

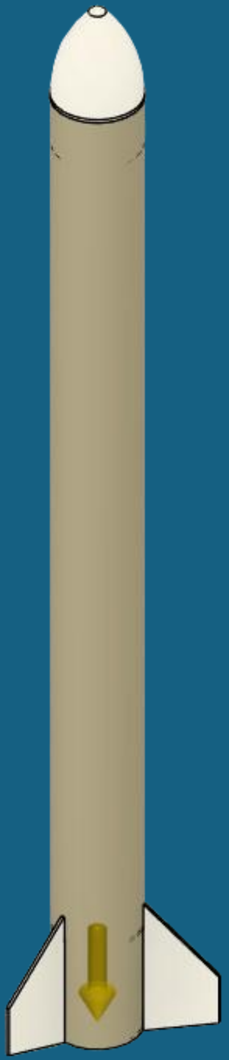


Engineering Skills & Readiness









- **Engineering Skills & Readiness — Key Takeaways**
  - Designed, built, and validated autonomous systems, aerospace structures, and lab-scale research rigs under real-world cost, safety, and material constraints.
  - Delivered measurable performance gains — up to **32% efficiency improvement** and **26% navigation accuracy boost** — through iterative design, PID tuning, and precision fabrication.
  - Skilled in **SolidWorks, MATLAB, ROS2, CAD/FEA, LIDAR, IMU, and embedded controls**, ready to solve complex mechanical challenges from concept to deployment.
  - Thrive in cross-disciplinary teams, leading projects from early design to tested, documented, and field-ready prototypes.

Let's Build Something Great

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